



School of Engineering and Architecture

Department of Electrical and Electronics Engineering

IOT-BASED AIR QUALITY MONITORING SYSTEM

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A proposal submitted in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Electrical and Electronics Engineering in the Department of Electrical and Electronics Engineering at Meru University of Science and Technology.

January, 2026

DECLARATION

I hereby declare that this project proposal is my original work except as cited in the references and has not been presented for the award of a degree in any other University.

Sign:Date:

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EG209/109705/22

This proposal has been submitted for examination with our approval as the University supervisors:

Sign: Date:

Supervisor: Mr. Job Kerosi

Sign: Date:

Supervisor: Madam Fiona Atieno

LIST OF ABBREVIATIONS

ESP32 - Espressif 32-bit Microcontroller

IoT – Internet of Things

DHT11 - Digital Humidity and Temperature Sensor

LPG - Liquefied Petroleum Gas

MQ - Metal Oxide Semiconductor Gas Sensor Series

MQ7 - Carbon Monoxide Gas Sensor

PM - Particulate Matter

PM_{2.5} – Particulate Matter with diameter ≤ 2.5 micrometers

PM₁₀ - Particulate Matter with diameter ≤ 10 micrometers

ML - Machine Learning

CO - Carbon Monoxide

RMSE - Root Mean Square Error

MAE - Mean Absolute Error

RF – Random Forest

IS – Isolation Forest

DT – Decision Tree

ABSTRACT

Air quality degradation and hazardous gas emissions pose significant challenges to public health, safety, and environmental sustainability. Urbanization and industrialization have increased the release of pollutants such as carbon oxides, methane, and particulate matter (PM_{2.5} and PM₁₀), while domestic environments remain at risk from leaks of liquefied petroleum gas (LPG) and other combustible gases. These pollutants are linked to respiratory illnesses, cardiovascular diseases, climate change, and fire hazards. Existing monitoring systems are often centralized, expensive, and inaccessible for real-time decision-making at the household or community level. This project proposes an IoT-based air quality monitoring system capable of real-time data acquisition, cloud-based storage, and predictive analytics. The system integrates low-cost MQ gas sensors, particulate matter sensors, and environmental sensors with an ESP32 microcontroller for data collection and processing. Data is transmitted to Firebase for secure storage, visualization, and alert dissemination through a web application. Predictive models are employed to forecast pollution levels, detect anomalies, and assess potential fire hazards, while an actuation and safety response mechanism ensures timely mitigation. The proposed system provides a scalable, resource-efficient solution for continuous air quality monitoring, early warning, and risk management across urban, domestic, agricultural, and healthcare environments.

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CHAPTER 1: INTRODUCTION

1.1 BACKGROUND OF STUDY

Air quality degradation has become a pressing environmental and public health concern globally. Rapid urbanization, increased industrial activities, transportation, and population growth have significantly elevated the release of harmful pollutants such as particulate matter (PM_{2.5} and PM₁₀), methane, carbon oxides, and volatile organic and inorganic compounds [1]. These pollutants are directly associated with respiratory illnesses, cardiovascular diseases, allergies, and climate impacts [1]. In addition to outdoor pollution, domestic environments face risks of hazardous gas leaks such as liquefied petroleum gas (LPG) and methane, which can lead to poisoning, fire outbreaks, and explosions [2].

Although government-operated monitoring stations exist in major cities, current air quality monitoring approaches remain centralized, expensive, and inaccessible to individuals at the household level [1]. This creates a gap in real-time localized monitoring that can help mitigate risks [3].

Recent advancements in Internet of Things (IoT), cloud computing, and machine learning present a promising opportunity to develop affordable and scalable monitoring systems [3]. The proposed IoT-Based Air Quality Monitoring System integrates low-cost sensors with cloud-based storage and real-time data visualization to deliver accessible environmental insights [4]; [5]. By leveraging MQ-series gas sensors, particulate matter sensors, temperature and humidity sensors (DHT11), and ESP32 microcontrollers, the system offers continuous surveillance of air pollutants and hazardous gases [4]; [5]. The integration of predictive analytics further enhances the capability to detect anomalies and forecast pollution trends [3]; [6]; [7].

The integration of an actuation and safety response system strengthens hazard mitigation at the household level [2]. Environmental factors such as temperature and humidity also influence sensor readings and system accuracy, making real-time calibration and compensation necessary [8]; [9]. The system, therefore, represents a timely initiative toward improving environmental health, public safety, and awareness by bridging the gap between centralized air quality monitoring and household-level accessibility [5].

1.2 PROBLEM STATEMENT

Air pollution and hazardous gas exposure remain persistent problems in urban and domestic environments due to increasing industrial activity, vehicle emissions, and the widespread use of gas-powered appliances. In many residential areas and small business premises, exposure to pollutants such as carbon monoxide and combustible gases often goes undetected until health symptoms, fires, or explosions occur, significantly increasing the risk of injury, property damage, and loss of life. Existing air quality and gas monitoring systems are largely centralized, expensive, and designed for regulatory or industrial applications rather than individual households or small enterprises. As a result, they are inaccessible to most users and fail to provide continuous, real-time, and location-specific information at the point of exposure, leaving individuals without timely awareness or early warning of deteriorating air quality or gas leakages in their immediate surroundings. The absence of a resource-efficient, real-time, and localized monitoring solution therefore creates a critical safety and public health gap, causing communities to remain reactive rather than preventive in managing air pollution and gas-related hazards. Addressing this gap requires an economical, scalable, and accessible monitoring system capable of detecting air pollutants and hazardous gases in real time, generating timely alerts, and enabling preventive response through automated actuation, which is the aim of the proposed IoT-Based Air Quality Monitoring System.

1.3 OBJECTIVES

Main objective

To design a scalable IoT-Based Air Quality Monitoring System with real-time detection, prediction, alerts, and actuation.

Specific objectives:

- a. To measure and log air pollutants (PM2.5, PM10, CO, LPG) along with temperature and humidity using low-cost sensors connected to an ESP32 microcontroller.
- b. To transmit collected data to a cloud platform for real-time visualization, predictive modeling of air quality trends, and threshold-based alert generation through a web interface.
- c. To implement automated actuation mechanisms that respond to detected hazards or predicted unsafe conditions, such as activating ventilation or triggering safety alerts.

1.4 JUSTIFICATION

Air quality monitoring solutions are predominantly confined to government-operated or industrial-scale installations, which limits accessibility for households and small businesses. Consequently, individuals are frequently exposed to invisible yet hazardous pollutants and gas leaks without timely detection or warning. An economical and scalable IoT-based Air Quality Monitoring System addresses this gap by enabling early detection and automated alerts that enhance public safety. It provides accessible, real-time environmental data that supports informed decision-making at the individual and community levels, thereby reducing exposure-related health risks. By promoting community awareness and integrating IoT infrastructure with machine learning analytics, the system enables continuous monitoring and early detection of air quality issues. It provides a practical, research-based solution for improving environmental health and public safety.

1.5 SCOPE

This project focuses on designing and implementing an IoT-based air quality monitoring system using low-cost gas sensors, particulate matter sensors, environmental sensors, actuators, and ESP32 microcontrollers. The scope includes real-time data collection, wireless transmission to a cloud platform (Firebase), web-based data visualization, safety response, and predictive modeling for anomaly detection and forecasting. The project will cover; hardware integration, firmware development, cloud connectivity, web application interface, alert systems, and evaluation of system performance. This project, however, does not cover nationwide monitoring or industrial certification standards. Instead, the focus remains on household, urban, agricultural, and healthcare environments where feasible and cost-effective deployment is needed.

CHAPTER 2: LITERATURE REVIEW

Air quality monitoring has become increasingly important due to rising pollution levels and associated health risks. Traditional reference-grade monitoring stations provide high measurement accuracy but are expensive and sparsely distributed, limiting their spatial coverage and accessibility. Bagkis et al. [1]; Kumar et al. [2] highlighted the technological evolution of low-cost sensor networks and emphasized the need for scalable, distributed monitoring solutions. Similarly, Garcia et al. [3] noted that centralized systems restrict accessibility and integration with modern digital infrastructures.

Recent research has therefore shifted toward low-cost sensors and IoT-based systems to improve real-time environmental monitoring. Concas et al. [4] surveyed low-cost outdoor air quality monitoring systems, discussing calibration challenges, sensor drift, and cross-sensitivity effects. Their work emphasized proper calibration strategies for reliable deployment at scale.

2.1 Technologies Used in Air Quality Monitoring

Low-cost sensors are increasingly preferred for distributed monitoring due to their affordability and compact size. Concas et al. [4] explained that sensor drift and environmental interference can affect measurement accuracy over time. Environmental conditions further impact sensor performance. Spinelle et al. [14] showed that temperature and humidity variations influence sensor sensitivity and stability, while Casari and Po [8] reported that high humidity inflates particulate matter readings due to hygroscopic growth.

To address these challenges, IoT architectures and machine learning techniques are widely used. Zaid et al. [10] developed an IoT-based low-cost sensor network capable of reliable wireless data transmission, real-time monitoring, and cloud storage. Ramesh et al. [11] reported that integrating sensors with microcontrollers and cloud dashboards enables real-time monitoring and early hazard detection. Garcia et al. [3] also highlighted the use of AI to improve predictive analytics and system automation.

Machine learning has improved sensor reliability and data accuracy. Tastan [6] applied machine learning for sensor calibration and reported improvements in measurement accuracy and stability. Kim and Seung-Hyun [7] reviewed a decade of machine learning methods for correcting sensor drift and noise, including Random Forest, SVM, and deep learning. Idir et al. [12] compared predictive models for urban air quality using mobile and fixed sensors, showing that data fusion enhances spatial resolution and forecasting accuracy.

2.2 Review of Existing Systems

Several studies have demonstrated the practical deployment of low-cost monitoring systems. Gatari et al. [13] conducted one of the earliest studies in Kenya using calibrated particulate matter sensors, validating their accuracy against reference instruments. Njeru et al. [14] deployed a PM2.5 sensor network in Mombasa, Kenya, reporting frequent exceedances of health-based air quality guidelines, indicating high exposure risks.

Community-integrated approaches improve monitoring effectiveness. Manshur et al. [15] implemented citizen-science air quality monitoring in a Kenyan informal settlement, showing both technical feasibility and social value. Bowyer et al. [16] demonstrated that participatory arts-based methods, such as theatre, music, murals, and school outreach, increase awareness and participation in air pollution studies.

Indoor and household-level monitoring systems are increasingly emphasized. Othman et al. [5] reviewed low-cost IoT-based indoor air quality monitoring systems, highlighting low power requirements and suitability for household deployment. Kumar et al. [2] developed a gas leakage detection system with automated alerts, showing the importance of safety response mechanisms in household monitoring.

Ramesh et al. [11] identified safety response mechanisms as a key component of air quality monitoring systems, enabling real-time hazard detection and mitigation.

2.3 Research Gaps

Despite advances in low-cost sensors, IoT platforms, and machine learning, several gaps remain. Household-level access to reliable air quality monitoring is still limited. Many current systems lack real-time alert mechanisms and predictive analytics integration. Community engagement strategies are often underutilized, reducing participation and trust. Finally, sensor performance can be affected by environmental conditions, and more work is needed to maintain accuracy under varying temperature and humidity. These gaps justify the development of an integrated solution combining distributed sensors, IoT communication, cloud-based data management, machine learning analytics, and community engagement to provide accessible and actionable air quality information.

CHAPTER 3: METHODOLOGY

3.1 SYSTEM REQUIREMENTS

1. ESP32 microcontroller.
2. DHT11 sensor - Temperature and humidity.
3. Mq7 sensor - carbon monoxide(CO).
4. MQ8 sensor – Hydrogen Gas
5. MQ5 sensor - LPG
6. Nova PM sensor Particulate matter.
7. Breadboard and stripboard.
8. 12 V battery and battery charger.
9. Powerbank.
10. Buzzer.
11. Relay.
12. Jumper wires.
13. Firebase
14. Arduino IDE
15. Visual studio code
16. GitHub
17. LCD Display.
18. Voltage regulator



Figure 1:Passive Buzzer

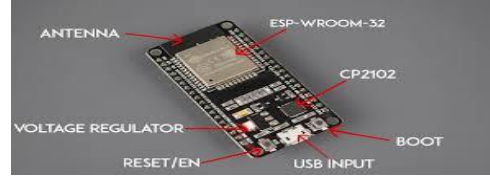


Figure 2: ESP32 Board



Figure 3: DHT11 Sensor



Figure 4: LCD Display



Figure 5: PM nova Sensor



Figure 6: MQ8 Sensor



Figure 7:MQ5 Sensor

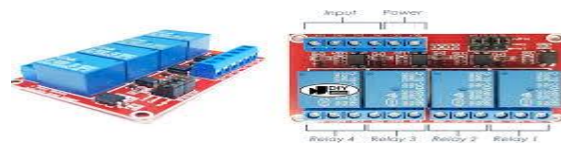


Figure 8: 4_way relay Channel



Figure 9: MQ7 Sensor



Figure 10: Voltage Regulator

3.2.3 Firmware Development

The program that will enable the ESP32 microcontroller to read values from the sensors, connect to the internet, perform necessary signal processing and package the data for transmission to the firebase database will be developed in Arduino IDE. The configuration keys for the database will be embedded into the program to enable seamless communication between the ESP32 and the firebase real-time database. This program will aid in data collection.

3.3 DATA COLLECTION

DHT11 sensor will be used for temperature and humidity monitoring, PM nova sensor will be used for particulate matter (PM_{2.5} and PM₁₀) monitoring, MQ5 will be used for LPG monitoring, MQ8 will be used for hydrogen monitoring, and MQ7 will be used for carbon monoxide (CO) monitoring.

The sensors will take readings of the respective air constituents and transmit the values in terms of a voltage signal to the analog pins of the ESP32 board, the signal will therefore be digitized inside the ESP32 using the analog to digital converters (ADCs) inside the ESP32 board. The values will be quantized within a 12-bit range of 0-4095 depending on the strength of the signal received and transmitted via internet to the firebase real-time database for storage. Through HyperText Transfer Protocol (HTTP), the data will be conveyed to the web application for real-time visualization.

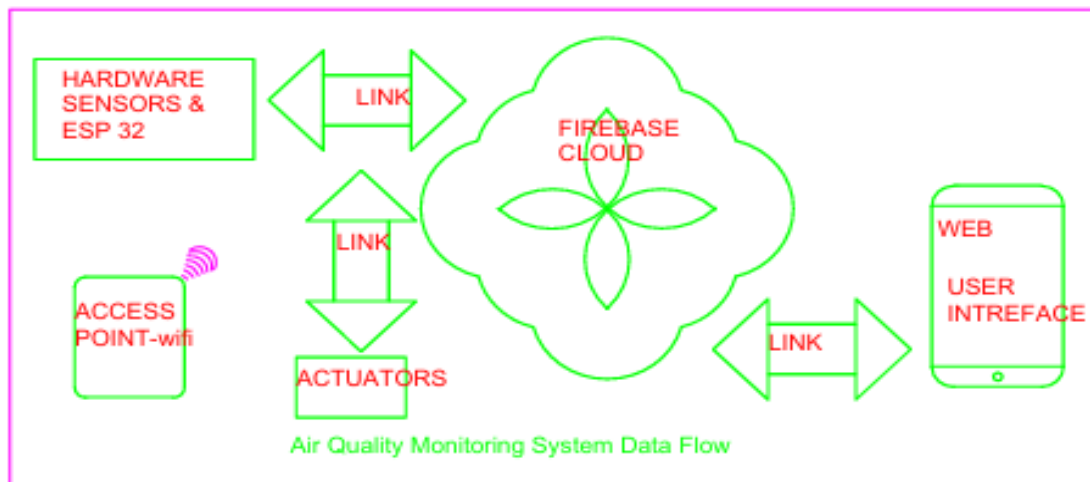


Figure 12: Flow of Data

The data collection exercise will span for a minimum 30 days and all of it will be stored in the firebase database where later it'll be accessed for analysis. The transfer of the data to the firebase database will occur after every 30 seconds. This is a design property to reduce the number of requests and responses (HTTP protocol) that the ESP32 has to make per unit time during transmission of the data to the cloud based database. This reduces network overhead and improves system stability ensuring the ESP32 doesn't overheat and prolongs its life and improves the overall performance of the system.

SYSTEM OPERATION

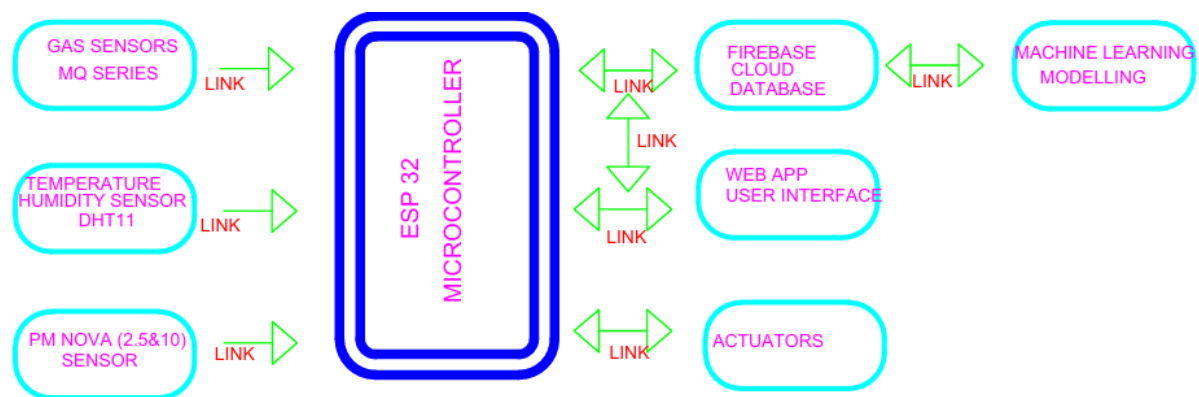


Figure 13: System Block Diagram

Data collected by the sensors will be passed to the ESP32 microcontroller for calibration, packaging and transmission to firebase for the purposes of storage. From firebase the data will be accessed by the web application for real-time visualization. The machine learning unit that is responsible for short term prediction and anomaly detection will access the data from firebase and perform necessary analytics. Depending on the results of the analytics alerts will be disseminated accordingly. The actuators will be activated or deactivated necessarily depending on the output of the analytics.

3.4 DATA ANALYSIS METHODS

Sensor data will be collected and stored in the Firebase database at a fixed interval of 30 seconds for a minimum duration of 30 days. The total number of data points will be estimated using the expression:

$$(\text{Days} \times \text{Hours_In_A_day} \times \text{Minutes_In_Hour} \times \text{Seconds_In_minute}) \div 30$$

For a 30-day data collection period, this results in approximately 86,400 data points per sensor. The collected dataset will be preprocessed to remove missing values, handle outliers, and ensure timestamp consistency before further analysis. The cleaned data will be used to train simple machine learning models aimed at short-term prediction and anomaly detection.

3.4.1 Machine Learning Modeling

The following machine learning models will be employed in this project:

I. Decision Tree (DT) - Decision Trees are supervised learning algorithms used for both classification and regression tasks. They operate by recursively splitting the input data based on feature values, forming a tree structure where internal nodes represent decision rules and leaf nodes represent output predictions. Decision Trees are easy to interpret, computationally efficient, and suitable for structured sensor data. However, they are sensitive to small variations in the dataset and are prone to overfitting. To reduce this effect, tree depth and splitting criteria will be controlled during training.

II. Random Forest (RF) - Random Forest is an ensemble learning technique that combines multiple Decision Trees to improve prediction accuracy and robustness. Each tree is trained on a random subset of the dataset, and final predictions are obtained through averaging (for regression) or majority voting (for classification). Random Forest models are less sensitive to noise and provide improved generalization compared to individual Decision Trees. Their main limitation is increased computational cost and reduced interpretability due to the ensemble structure.

III. Isolation Forest (IF)

Isolation Forest is an unsupervised learning algorithm designed for anomaly detection. It works by isolating data points through random partitioning, where anomalies require fewer splits to be isolated compared to normal data points. This model will be used to detect abnormal pollution patterns such as sudden gas concentration spikes that deviate from normal operating conditions.

3.4.2 Models' Training and Deployment

The collected data will be subdivided in the ratio 80:20 where 80% of the data will be used for the models' training and the remaining 20% will be used for testing the models. The models' training will be done locally on the PC and will be run either locally on the PC or remotely on google cloud. The models cannot be run inside the ESP32 board because it's limited in hardware resources. The predicted values (random Forest, RD) will be displayed on the web dashboard and incase of anomaly detection (Isolation forest, IF) alerts will be triggered on the web dashboard for action.

3.4.3 Model Evaluation

- a) Mean Absolute Error (MAE) - MAE measures the average magnitude of errors between the predicted values and the actual sensor readings, ignoring the direction of the errors. It is calculated as the arithmetic mean of the absolute differences between predicted and actual values.

The diagram shows the MAE formula: $MAE = \frac{1}{n} \sum |y - \hat{y}|$. Annotations include: a blue box around $\frac{1}{n}$ with the text "Divide by the total number of data points"; a green box around y with the text "Actual output value"; an orange box around $\hat{y}$ with the text "Predicted output value"; a bracket under $|y - \hat{y}|$ with the text "The absolute value of the residual"; and the word "Sum of" below the summation symbol $\sum$.

- b) Root Mean Square Error (RMSE) - RMSE measures the square root of the average squared differences between predicted and actual values:

$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$

$\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n$ are predicted values

$y_1, y_2, \dots, y_n$ are observed values

n is the number of observations

Unlike MAE, RMSE penalizes larger errors more heavily, making it more sensitive to outliers. Together with MAE, RMSE provides a comprehensive view of the model's prediction accuracy.

c) Confusion Matrix

Isolation Forest will be evaluated using a confusion matrix to allow for a detailed assessment of its ability to detect anomalies in air quality. From the confusion matrix, metrics such as accuracy, precision, recall, and F1-score will be derived. This evaluation ensures that the Isolation Forest reliably detects abnormal pollution events, such as sudden spikes in gas concentration, while minimizing false alarms.

3.4.4 SYSTEM FLOWCHART

The flowchart shows flow of the system's logic. Data is collected by sensors and ESP handles necessary processing for transmission to firebase, from firebase data is visualized on the web app, the machine learning unit (Python server) handles the short term prediction and anomaly detection and from the results alerts are disseminated and necessary actuation mechanisms activated.

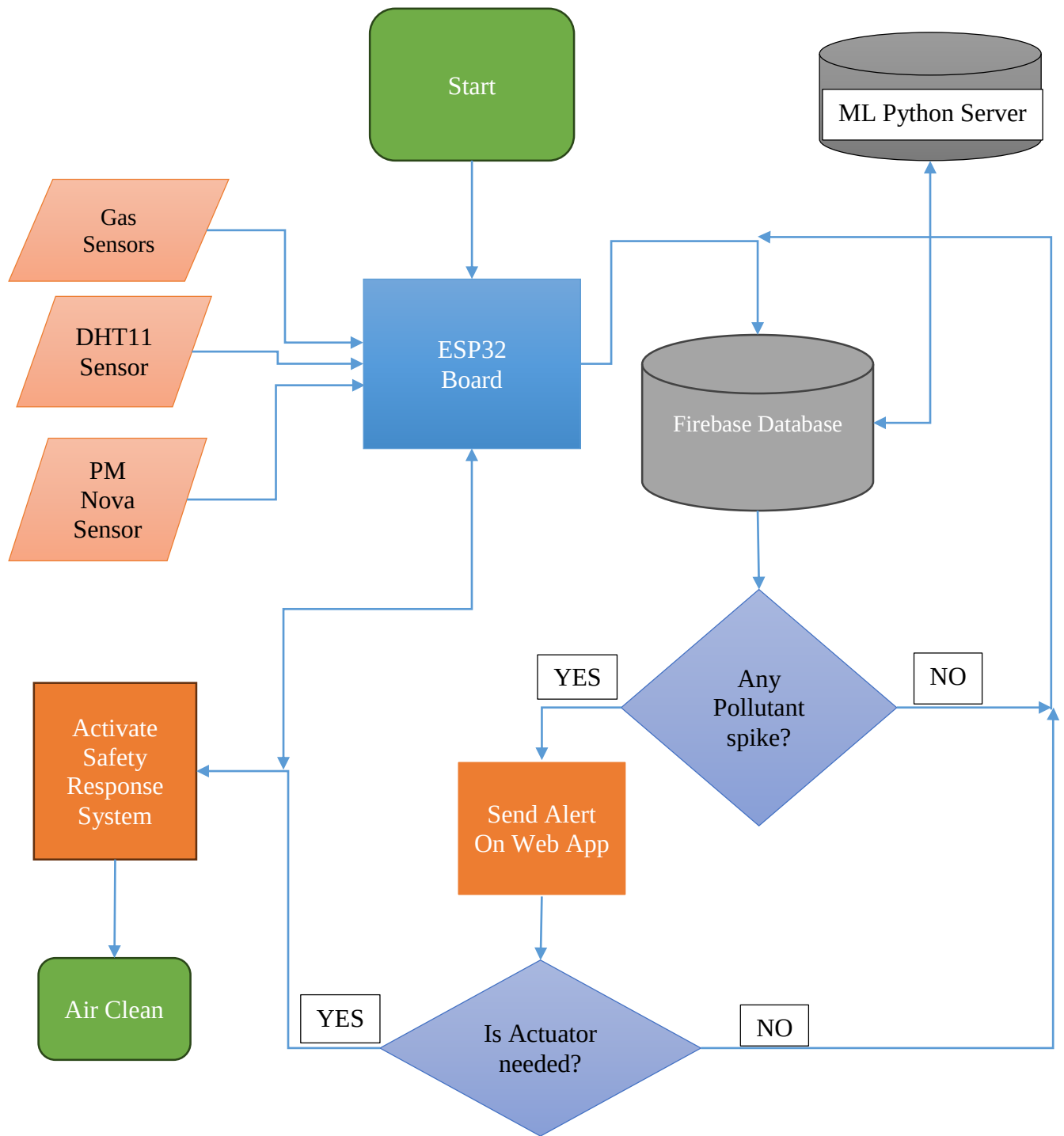


Figure 14: System Flow Chart

3.5 PROTOTYPE TESTING

The developed prototype will be tested under both controlled and real-world operating conditions to verify system functionality and performance. Controlled testing will be carried out using simulated pollutant sources to evaluate sensor response, threshold detection, and alert triggering behavior. These tests will be used to confirm correct sensor interfacing, data acquisition, and cloud data transmission. Real-world testing will be conducted in a domestic environment to assess system reliability under normal environmental variations such as temperature changes, ventilation differences, and background pollution levels. Key performance parameters to be evaluated will include sensor responsiveness, alert latency, cloud data update consistency, and web dashboard functionality. The results of these tests will be used to validate that the system meets its intended functional and operational requirements.

3.6 ETHICAL AND SAFETY CONSIDERATIONS

1. Gas sources used during controlled testing are handled in well-ventilated environments to minimize the risk of accidental exposure.
2. Testing procedures are designed to ensure that no individuals are exposed to hazardous gas concentrations at any stage of system evaluation.
3. The system collects only environmental sensor data for research and performance evaluation purposes. No personal or sensitive user data is recorded, and access to stored data is restricted to authorized users.

EXPECTED OUTCOMES

Real-Time Monitoring of Air Pollutants - The system will continuously monitor carbon monoxide, combustible gases (LPG and hydrogen), and particulate matter using MQ-series gas sensors and particulate sensors interfaced with the ESP32 microcontroller. Sensor data will be sampled locally and transmitted to the cloud database every thirty seconds, enabling near real-time observation of pollutant concentration variations in the monitored environment through a web-based dashboard.

Anomaly Detection, Short-Term Prediction, and Automated Alerts - Collected sensor data will be analyzed to identify abnormal patterns such as sudden gas concentration spikes or values exceeding predefined safety thresholds. Machine learning models including Decision Tree (DT), Random Forest (RF), and Isolation Forest (IF) will be applied to historical cloud data to perform short-term prediction of pollutant levels. When abnormal conditions or threshold exceedances are detected, the system will automatically trigger alerts through the cloud platform and display notifications on the web application interface, enabling early detection of potential gas leaks or unsafe air quality events.

Enhanced Environmental Awareness - By providing continuous air quality data, historical trends, and alert notifications, the system will improve user awareness of indoor or localized outdoor environmental conditions. Users will observe how pollutant levels change over time and relate these variations to activities such as cooking, fuel usage, or ventilation, supporting informed decision-making and safer environmental practices based on measured data.

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PROJECT WORKPLAN

The project will span for 8 months.

Project Gantt Chart.

ACTIVITY	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR
Project topic choice								
Introduction and background study on the topic								
Literature review and proposal writing								
Proposal presentation and submission								
Component procurement and prototype building								
Web, database, and firmware development.								
Prototype deployment, data collection, processing, analysis and machine learning modelling								
Project presentation and submission								

Figure 15: Project Gantt chart

PROJECT BUDGET

Serial No	Component	DESCRIPTION	Unit cost	Quantity	Total
1	ESP 32	DOIT ESP32 DEVKIT V1	1500	1	1500
2	MQ-SERIES	MQ-135 / MQ-7 gas sensors (analog modules)	400	4	1600
3	NOVA PM	SDS011 PM2.5/PM10 particulate sensor (UART)	2300	1	2300
4	BATTERY	12 V Li-ion / lead-acid battery, 2–5 Ah typical	1800	1	1800
5	BATTERY CHARGER	charger module	1200	1	1200
6	PSU	AMS1117-3.3 voltage regulator (3.3 V output).	400	1	400
7	JUMPERS	20 cm male-to-male jumper wires. (common prototyping set)	250	2	500
8	STRIP BOARD	Standard perfboard, <i>proto assembly</i>	100	2	200
9	BREAD BOARD	Standard 830-tie point breadboard.	200	1	200
10	ENCLOSURE	wooden box	1000	1	1000
11	INTERNET	Wi-Fi data connection plan (monthly)	1000	2	2000
12	DHT11	Temp/humidity sensor module (5 V)	500	1	500
13	ACCESS POINT	Wi-Fi router	2500	1	2500
14	POWERBANK	10 000 mAh USB power bank	1200	1	1200
15	Miscellaneous	Screws, headers, mounts.	400	1	400
16	LEDS	5 mm LEDs (e.g., red, green, blue).	10	10	100
17	BUZZER	5 V active piezo buzzer.	2	20	40
18	PUSH BUTTON	Standard tactile push button/switch	1	20	20
19	RESISTORS	Assorted: 10 k Ω , 4.7 k Ω , 1 k Ω , 220 Ω	10	20	200
20	CAPACITORS	Decoupling and bulk: 100 μF , 10 μF , 0.1 μF	20	10	200
				TOTAL	17860

